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[1]

$$(1) \sum_{n=2}^{\infty} \frac{i^n}{\ln^n} \quad \ln^n < n$$

$$\Rightarrow \sum_{n=2}^{\infty} \left| \frac{i^n}{\ln^n} \right| = \sum_{n=2}^{\infty} \frac{1}{\ln^n} > \sum_{n=2}^{\infty} \frac{1}{n} = \infty$$

so $\sum_{n=2}^{\infty} \frac{i^n}{\ln^n}$ not absolute converge

$$\sum_{n=2}^{\infty} \frac{i^n}{\ln^n} = \sum_{k=0}^{\infty} - \left[\left(\frac{1}{\ln^{(4k+2)}} - \frac{1}{\ln^{(4k+4)}} \right) + i \left(\frac{1}{\ln^{(4k+3)}} - \frac{1}{\ln^{(4k+5)}} \right) \right]$$

$$\textcircled{1} \sum_{k=0}^{\infty} - \left[\frac{1}{\ln^{(4k+2)}} - \frac{1}{\ln^{(4k+4)}} \right] = \sum_{m=1}^{\infty} \frac{(-1)^m}{\ln^{(2m)}} \Rightarrow \text{converge}$$

$$\textcircled{2} \sum_{k=0}^{\infty} -i \left[\frac{1}{\ln^{(4k+3)}} - \frac{1}{\ln^{(4k+5)}} \right] = \sum_{m=1}^{\infty} i \frac{(-1)^m}{\ln^{(2m+1)}} \Rightarrow \text{converge}$$

$$\Rightarrow \sum_{n=2}^{\infty} \frac{i^n}{\ln^n} \text{ converge}$$

$$(2) \sum_{n=1}^{\infty} \frac{i^n}{n}$$

$$\sum_{n=1}^{\infty} \left| \frac{i^n}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n} = \infty, \text{ not absolute converge}$$

$$\sum_{n=1}^{\infty} \frac{i^n}{n} = \sum_{n=1}^{\infty} \left(-\frac{1}{4n-2} + \frac{1}{4n} \right) + i \left(\frac{1}{2n-3} - \frac{1}{2n-1} \right)$$

$$\textcircled{1} \sum_{n=1}^{\infty} \left(-\frac{1}{4n-2} + \frac{1}{4n} \right) = \sum_{n=1}^{\infty} \frac{1-1)^n}{2n} \Rightarrow \text{converge}$$

$$\textcircled{2} \sum_{n=1}^{\infty} i \left(\frac{1}{2n-3} - \frac{1}{2n-1} \right) = \sum_{n=1}^{\infty} i \frac{1-1)^{n-1}}{2n-1} \Rightarrow \text{converge}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{i^n}{n} \text{ converge}$$

[2]

$$(1) \sum_{n=1}^{\infty} z^{n!}$$

$$\text{For } |z| > 1, |z|^{n!} > |z| \Rightarrow \sum_{n=1}^{\infty} |z| = \infty$$

$$\Rightarrow |z| > 1 \Rightarrow \text{diverge}$$

$$\text{For } |z| < 1, |z|^{n!} < |z| \Rightarrow \sum_{n=1}^{\infty} |z| < \infty$$

$$\Rightarrow |z| < 1 \Rightarrow \text{converge}$$

Convergence region : $|z| < 1$

$$(2) \sum_{n=1}^{\infty} \left(\frac{z}{1+z} \right)^n$$

$$\Rightarrow \text{converge when } \left| \frac{z}{1+z} \right| < 1$$

$$\Rightarrow \left| \frac{x+yi}{1+x+yi} \right| < 1$$

$$\Rightarrow x > -\frac{1}{2} \Rightarrow \operatorname{Re} z > -\frac{1}{2}$$

$$(3) \sum_{n=1}^{\infty} (-1)^n (z^2 + 2z + 2)^n$$

$$\Rightarrow |z^2 + 2z + 2| < 1$$

$$(4) \sum_{n=1}^{\infty} z^n \sin \frac{z}{3^n}$$

$$\begin{aligned} \sin z &= \frac{e^{iz} - e^{-iz}}{2i} = \frac{1}{2i} (e^{-y+ix} - e^{y-ix}) \\ &= \frac{1}{2i} [e^{ix}(e^{-y} - e^y) + e^y(e^{ix} - e^{-ix})] \\ &= \frac{1}{2i} e^{ix} (e^{-y} - e^y) + e^y \sin x \end{aligned}$$

$$\text{since } z \rightarrow 0 \quad z = x+iy \Rightarrow e^y \sin x \sim x$$

$$\Rightarrow \frac{1}{2i} e^{ix} (e^{-y} - e^y) \sim \frac{1}{2i} (-2y) = yi$$

$$\Rightarrow \sum_{n=1}^{\infty} z^n \left[\frac{1}{2i} e^{i \frac{x}{3^n}} (e^{-\frac{y}{3^n}} - e^{\frac{y}{3^n}}) \right]$$

$$\text{converge if } \sum_{n=1}^{\infty} z^n \frac{y}{3^n} \text{ i converge}$$

$$\Rightarrow \sum_{n=1}^{\infty} z^n e^{\frac{y}{3^n}} \sin \frac{x}{3^n} \text{ converge if } \sum_{n=1}^{\infty} z^n \frac{x}{3^n} \text{ converge}$$

Since

$$\sum_{n=1}^{\infty} z^n \sin \frac{x}{3^n} = \sum_{n=1}^{\infty} z^n \left[\frac{1}{2i} e^{i \frac{x}{3^n}} (e^{-\frac{y}{3^n}} - e^{\frac{y}{3^n}}) + e^{\frac{y}{3^n}} \sin \frac{x}{3^n} \right]$$

$$\Rightarrow \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n y i, \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n x \text{ converge}$$

$$\Rightarrow \forall z \in \mathbb{C}, \sum_{n=1}^{\infty} z^n \sin \frac{x}{3^n} \text{ converge}$$

[3]

$$(1) \sum_{n=1}^{\infty} \frac{1}{n^n} z^n$$

$$R = \lim_{n \rightarrow \infty} |n^n|^{\frac{1}{n}} = \infty$$

$$(2) \sum_{n=1}^{\infty} \frac{1}{2^n \cdot n^n} z^n$$

$$R = \lim_{n \rightarrow \infty} |(2n)^n|^{\frac{1}{n}} = \infty$$

$$(3) \sum_{n=1}^{\infty} \frac{n!}{n^n} z^n$$

$$R = \lim_{n \rightarrow \infty} \frac{\frac{n!}{n^n}}{\frac{(n+1)!}{(n+1)^{n+1}}} = \lim_{n \rightarrow \infty} \left(\frac{1}{n+1}\right) \left(1 + \frac{1}{n}\right)^n (n+1) = e$$

$$(4) \sum_{n=1}^{\infty} \frac{(-1)^n}{2^{2n} (n!)^2} z^n$$

$$R = \lim_{n \rightarrow \infty} (\quad) \text{ since } (-1)^n$$

$$\Rightarrow R = \infty$$

$$15) \sum_{n=1}^{\infty} n^{\ln n} z^n$$

$$R = \lim_{n \rightarrow \infty} (n^{-\ln n})^{\frac{1}{n}}$$

$$= \lim_{n \rightarrow \infty} (e^{\ln n \cdot \ln n^{-1}})^{\frac{1}{n}}$$

$$= \lim_{n \rightarrow \infty} e^{-\frac{(\ln n)^2}{n}}$$

$$= 1$$

$$16) \sum_{n=1}^{\infty} \frac{1}{2^{2n}} z^{2n}$$

$$R = \lim_{n \rightarrow \infty} (2^{2n})^{\frac{1}{n}} \cdot \frac{1}{2} = 2$$

$$17) \sum_{n=1}^{\infty} \frac{\ln^n n}{n!} z^n$$

$$R = \lim_{n \rightarrow \infty} \frac{\frac{\ln^n n}{n!}}{\frac{\ln^{n+1} (n+1)}{(n+1)!}} = \lim_{n \rightarrow \infty} (n+1) \frac{n \ln^n n}{(n+1) \ln^{n+1} (n+1)} = \infty$$

$$18) \sum_{n=1}^{\infty} (1 - \frac{1}{n})^n z^n$$

$$R = \lim_{n \rightarrow \infty} \left(\left(1 - \frac{1}{n} \right)^{-n} \right)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{n}{n-1} = 1$$

[4]

$$(1) 1 - z^2 = 1 - [(z-1)+1]^2 = -2(z-1) - (z-1)^2$$

$$\Rightarrow \text{radius } R = \infty \Rightarrow |z-1| < \infty$$

$$(2) \sin z = (-1)^n \sin(z - n\pi)$$

$$= (-1)^n \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} (z - n\pi)^{2k+1}$$

$$= \sum_{k=0}^{\infty} \frac{(-1)^{n+k}}{(2k+1)!} (z - n\pi)^{2k+1}$$

$$\text{radius } R = \infty \Rightarrow |z - n\pi| < \infty$$

$$(3) \frac{1}{1+z+z^2} \text{ at } z=0$$

$$\Rightarrow z^2 + z + 1 = 0 \Rightarrow z_1 = \frac{-1 + \sqrt{3}i}{2}, z_2 = \frac{-1 - \sqrt{3}i}{2}$$

$$\frac{1}{1+z+z^2} = \frac{1}{z_1 - z_2} \left(\frac{1}{z - z_1} - \frac{1}{z - z_2} \right)$$

$$= \frac{-1}{\sqrt{3}i} \left(\frac{1}{z_1} \frac{1}{1 - \frac{z}{z_1}} - \frac{1}{z_2} \frac{1}{1 - \frac{z}{z_2}} \right)$$

$$= \frac{1}{\sqrt{3}} \left[\frac{1}{z_1} \sum_{n=0}^{\infty} \left(\frac{z}{z_1} \right)^n - \frac{1}{z_2} \sum_{n=0}^{\infty} \left(\frac{z}{z_2} \right)^n \right]$$

$$= \frac{1}{\sqrt{3}} \sum_{n=0}^{\infty} z^n \left(e^{-\frac{2}{3}(n+1)\pi i} - e^{-\frac{2}{3}(n+1)\pi i} \right)$$

$$= -\frac{2}{\sqrt{3}} \sum_{n=0}^{\infty} \frac{z^n}{2i} (e^{-\frac{2}{3}(n+1)\pi i} - e^{\frac{2}{3}(n+1)\pi i})$$

$$= \sum_{n=0}^{\infty} \frac{\sin \frac{2}{3}(n+1)\pi}{\sin \frac{2}{3}\pi} z^n$$

radius : $R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = 1 \Rightarrow |z| < 1$

(4) $\frac{\sin z}{1-z}$ at $z=0$

$$\frac{\sin z}{1-z} = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} z^{2k+1} \cdot \sum_{m=0}^{\infty} z^m$$

$$= \sum_{n=1}^{\infty} \left[\sum_{2k+1+m=n} \frac{(-1)^k}{(2k+1)!} \right] z^n$$

$$= \sum_{n=0}^{\infty} \left[\sum_{k=0}^{\frac{n-1}{2}} \frac{(-1)^k}{(2k+1)!} \right] z^n$$

radius $R=1 \Rightarrow |z| < 1$

(5) $\exp\left(\frac{1}{1-z}\right)$, at $z=0 \Rightarrow R=1$

$$e^{\frac{1}{1-z}} = \sum_{k=0}^{\infty} a_k z^k, \quad \frac{e^{\frac{1}{1-z}}}{(1-z)^2} = \sum_{k=1}^{\infty} k a_k z^{k-1}$$

$$\sum_{k=0}^{\infty} a_k z^k = \sum_{n=0}^{\infty} (n+1) a_{n+1} z^n - 2 \sum_{n=1}^{\infty} n a_n z^n + \sum_{n=2}^{\infty} (n-1) a_{n-1} z^n$$

$$\Rightarrow a_{n+1} = \frac{2n+1}{n+1} a_n - \frac{n-1}{n+1} a_{n-1}, \quad n \geq 2$$

Since recursion $a_0 = e$

$$\Rightarrow e^{\frac{1}{1-z}} = e \left(1 + z + \frac{3}{2} z^2 + \frac{13}{6} z^3 + \dots \right)$$
$$= \sum_{n=0}^{\infty} \frac{1}{n!} \frac{d^n}{dz^n} (z^{n+1} e^z) \Big|_{z=1} z^n$$
$$\Rightarrow |z| < 1$$

[5]

$$(1) \sum_{n=0}^{\infty} \frac{1}{2n+1} z^{2n+1}$$

$$\text{suppose } f(z) = \sum_{n=0}^{\infty} \frac{z^{2n+1}}{2n+1}$$

$$f'(z) = \sum_{n=0}^{\infty} z^{2n}$$

$$= \frac{1}{1-z^2}$$

$$= \frac{1}{2} \left(\frac{1}{z-1} - \frac{1}{z+1} \right)$$

$$\Rightarrow f(z) = \frac{1}{2} \ln \frac{1+z}{1-z} = \tan^{-1} z$$

$$R=1 \Rightarrow |z| < 1, f(0) = 0$$

$$(2) \sum_{n=0}^{\infty} \frac{1}{(2n)!} z^{2n}$$

$$\Rightarrow \sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!} = \frac{1}{2} \left[\sum_{n=0}^{\infty} \frac{z^n}{n!} + \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} z^n \right]$$

$$= \frac{1}{2} (e^z + e^{-z})$$

$$= \cosh z$$

$$\Rightarrow R = \infty \Rightarrow |z| < \infty$$

$$(3) \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{(n+m)!}{n! m!} \left(\frac{z}{2}\right)^{n+m}$$

$$\sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k \sum_{n=0}^k \frac{k!}{n! (k-n)!} = \sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k \left(\sum_{n=0}^k \binom{k}{n}\right) = \sum_{k=0}^{\infty} 2^k \left(\frac{z}{2}\right)^k$$

$$\Rightarrow \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{(n+m)!}{n! m!} \left(\frac{z}{2}\right)^{n+m} = \frac{1}{1-z}$$

$$\Rightarrow |z| < 2 \cap \operatorname{Re} z < 1$$

$$(4) \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \sum_{p=0}^{\infty} \frac{(n+m+p)!}{n! m! p!} \left(\frac{z}{3}\right)^{n+m+p}$$

Same as (3)

$$\sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \sum_{p=0}^{\infty} \frac{(n+m+p)!}{n! m! p!} \left(\frac{z}{3}\right)^{n+m+p} = \frac{1}{1-z}$$

$$\Rightarrow |z| < 3 \cap \operatorname{Re} z < \frac{3}{2} \cap |z-2| < 1$$

[6]

$$(1) \frac{1}{z^2(z-1)} \text{ around } z=1$$

$$\Rightarrow \frac{1}{z^2(z-1)} = \sum_{n=-\infty}^{\infty} a_n (z-1)^n$$

$$a_n = \frac{1}{2\pi i} \int_{|z-1|=\rho} \frac{1}{z^2(z-1)} \cdot \frac{dz}{(z-1)^{n+1}}$$
$$= \frac{1}{2\pi i} \int_{|z-1|=\rho} \frac{1}{z^2} \cdot \frac{dz}{(z-1)^{n+2}}$$

$$\textcircled{1} \quad n \leq -2, \quad a_n = 0$$

$$\textcircled{2} \quad n = -1, \quad a_n = \frac{1}{z^2} \Big|_{z=1} = 1$$

$$\textcircled{3} \quad n \geq 0$$

$$a_n = \frac{1}{(n+1)!} \frac{d^{n+1}}{dz^{n+1}} (z^{-2}) \Big|_{z=1}$$

$$= \frac{1}{(n+1)!} (-2)(-3) \cdots [-(n+2)] z^{-(n+3)} \Big|_{z=1}$$

$$= (n+2)(-1)^{n+1}$$

$$\Rightarrow \frac{1}{z^2(z-1)} = \sum_{n=0}^{\infty} (n+2)(-1)^{n+1} (z-1)^n + (z-1)^{-1}$$

$$= \sum_{n=-1}^{\infty} (-1)^{n+1} (n+2) (z-1)^n$$

(2) area $1 < |z| < \infty$

$$\frac{1}{z^2(1-z)} = \frac{1}{z^3} \cdot \frac{1}{1-\frac{1}{z}} = \frac{1}{z^3} \sum_{n=0}^{\infty} \left(\frac{1}{z}\right)^n = \sum_{n=3}^{\infty} z^{-n}$$

13) $\frac{1}{z^2-3z+2}$, $1 < |z| < 2$

$$\frac{1}{z^2-3z+2} = \frac{1}{z-2} - \frac{1}{z-1}$$

$$= -\frac{1}{2(1-\frac{z}{2})} - \frac{1}{z(1-\frac{1}{z})}$$

$$= -\frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{z}{2}\right)^n - \sum_{n=0}^{\infty} \frac{1}{z^{n+1}}$$

$$= -\sum_{n=-1}^{-\infty} z^n - \sum_{n=0}^{\infty} \frac{z^n}{z^{n+1}}$$

(4) $\frac{1}{z^2-3z+2}$, $2 < |z| < \infty$

$$\frac{1}{z^2-3z+2} = \frac{1}{z-2} - \frac{1}{z-1}$$

$$= \frac{1}{z} \left(\frac{1}{1-\frac{2}{z}} - \frac{1}{1-\frac{1}{z}} \right)$$

$$= \sum_{n=1}^{\infty} (2^{n-1} - 1) z^{-n}$$

(5) $\frac{(z-1)(z-2)}{(z-3)(z-4)}$, $3 < |z| < 4$

$$\begin{aligned}
 \frac{(z-1)(z-2)}{(z-3)(z-4)} &= (z-1)(z-2) \left(-\frac{1}{4(1-\frac{z}{4})} - \frac{1}{z(1-\frac{3}{z})} \right) \\
 &= (z-1)(z-2) \left[-\frac{1}{4} \sum_{n=0}^{\infty} \left(\frac{z}{4}\right)^n - \frac{1}{z} \sum_{n=0}^{\infty} \left(\frac{3}{z}\right)^n \right] \\
 &= 1 - \frac{3}{2} \sum_{n=0}^{\infty} \frac{z^n}{2^{2n}} - 2 \sum_{n=1}^{\infty} \frac{z^n}{3^{n+1}}
 \end{aligned}$$

1b) $\frac{(z-1)(z-2)}{(z-3)(z-4)}$, $4 < |z| < \infty$

$$\begin{aligned}
 \frac{(z-1)(z-2)}{(z-3)(z-4)} &= (z-1)(z-2) \left(\frac{1}{z-4} - \frac{1}{z-3} \right) \\
 &= \left(z-3 + \frac{2}{z} \right) \left(\sum_{n=0}^{\infty} \left(\frac{4}{z}\right)^n - \sum_{n=0}^{\infty} \left(\frac{3}{z}\right)^n \right) \\
 &= 1 + \sum_{n=1}^{\infty} (3 \cdot 2^{2n-1} - 2 \cdot 3^{n-1}) z^{-n}
 \end{aligned}$$

[7]

(1) $\frac{1}{z^2 + a^2}$, $a \neq 0$

$$\frac{1}{z^2 + a^2} = \frac{1}{2ai} \left[\frac{1}{z-ai} - \frac{1}{z+ai} \right]$$

$\Rightarrow z = \pm ai$ first order pole

$\Rightarrow \lim_{z \rightarrow \infty} \frac{1}{z^2 + a^2} = 0 \Rightarrow \infty$ removable singularity

$$12) \frac{\cos az}{z^2}$$

$$\Rightarrow \frac{\cos az}{z^2} = \frac{\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n}}{z^2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n-2}$$

0, pole of order 2

∞ , essential singularity

$$13) \frac{\cos az - \cos bz}{z^2}, \quad a^2 \neq b^2$$

$$\begin{aligned} \frac{\cos az - \cos bz}{z^2} &= \frac{1}{z^2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} [a^{2n} - b^{2n}] z^{2n} \\ &= \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)!} [a^{2n} - b^{2n}] z^{2n-2} \end{aligned}$$

0, removable singularity

∞ , essential singularity

$$14) \frac{\sin z}{z^2} - \frac{1}{z}$$

$$\frac{\sin z - z}{z^2} = \frac{1}{z^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n+1} = \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n-1}$$

0, removable singularity

∞ , essential singularity

$$15) \cos \frac{1}{\sqrt{z}}$$

$$\cos \frac{1}{\sqrt{z}} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (\sqrt{z})^{2n} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^n$$

0, essential singularity

$$16) \frac{\sqrt{z}}{\sin \sqrt{z}} = \sqrt{z} \left(\frac{1}{\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^n \cdot \sqrt{z}} \right) = \frac{1}{\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^n}$$

0, removable singularity

$(n\pi)^2$, pole of order 1, $n=1, 2, \dots$

∞ , non-isolated singularity

$$17) \frac{1}{(z-1) \ln z}$$

$$\lim_{z \rightarrow 1} \frac{z-1}{\ln z - \ln 1} = \lim_{z \rightarrow 1} z = 1$$

$$\Rightarrow \frac{1}{(z-1) \ln z} = \frac{1}{(z-1)^2} \cdot \frac{z-1}{\ln z}$$

when $\ln z|_{z=1} = 0 \Rightarrow 1$, pole of order 2

otherwise, 1, pole of order 1

$$18) \int_0^z \frac{\sinh \sqrt{s}}{\sqrt{s}} ds$$

$$\int_0^z \frac{\sinh \sqrt{s}}{\sqrt{s}} ds = \int_0^z \sum_{k=1}^{\infty} \frac{s^{k-1}}{(2k-1)!} ds$$

$$= \sum_{k=1}^{\infty} \frac{z^k}{k(2k-1)!}$$

∞ , essential singularity